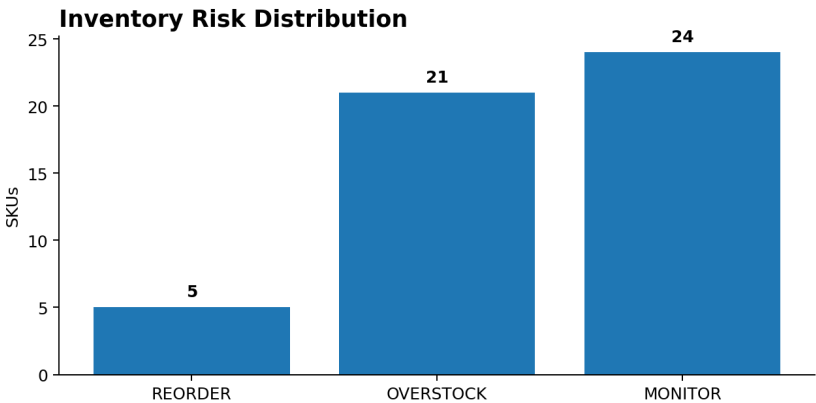
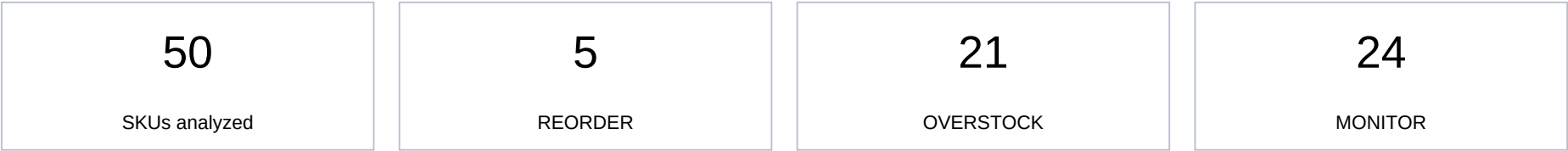


PROJECT FORESIGHT

Demand & Inventory Intelligence — Executive Readout

NorthBay Living | Data Science & Analytics



Executive interpretation

The analysis identifies three operational action groups: 5 SKUs flagged for REORDER, 21 for OVERSTOCK, and 24 for MONITOR. These classifications are generated from the project’s forecast and inventory rules.

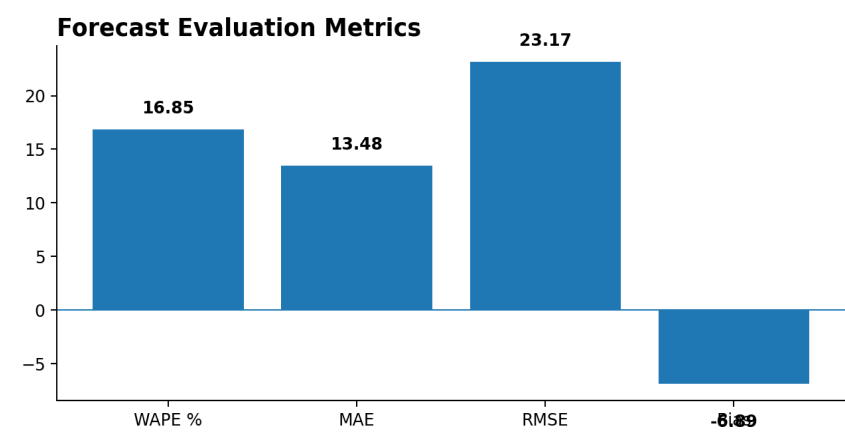
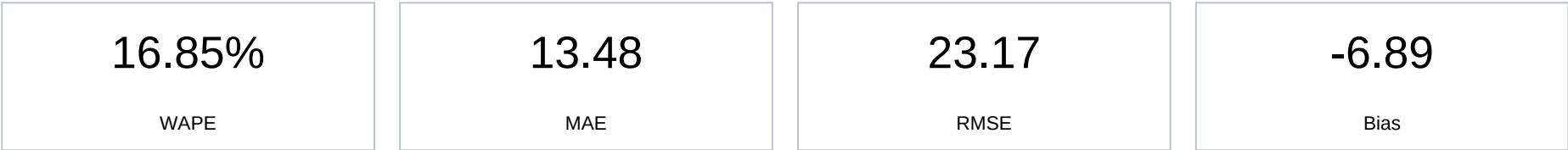
Modeled combined inventory exposure

■ **15.16M**

This is a modeled exposure estimate, not realized loss or guaranteed savings.

01 | Forecast Performance

Model evaluation on the project's final evaluation window.



What the metrics show

The primary project metric is WAPE, measured at **16.85%**. MAE is **13.48** units and RMSE is **23.17** units.

Bias is **-6.89**, indicating that model forecasts were lower than actual demand on average over the evaluation window. This should be considered when interpreting reorder decisions.

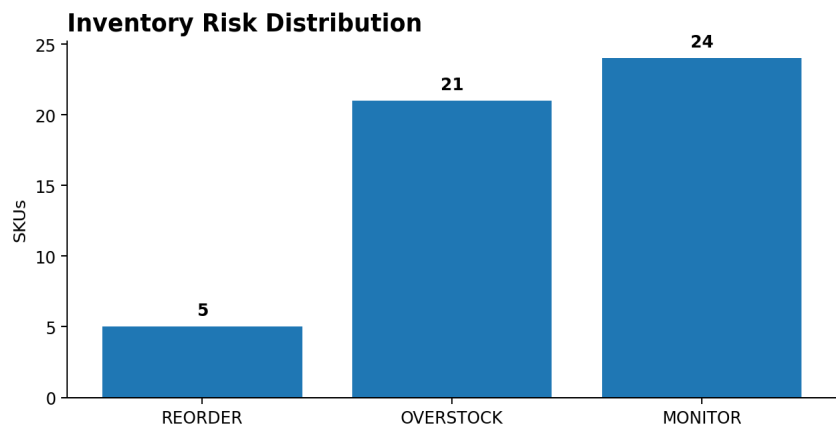
Model used: Random Forest regression with lag, rolling, calendar, SKU, price and promotion features.

Important: The readout reports the observed backtest metrics; it does not claim that the model will achieve the same accuracy on future unseen periods.

02 | Inventory Risk & Decision Logic

Risk scoring compares forecast demand with available inventory and lead-time requirements.

REORDER	OVERSTOCK	MONITOR
Available stock is below estimated lead-time demand.	Current stock is above the forward demand threshold used by the project.	Neither stockout nor overstock condition is triggered.



Risk mix

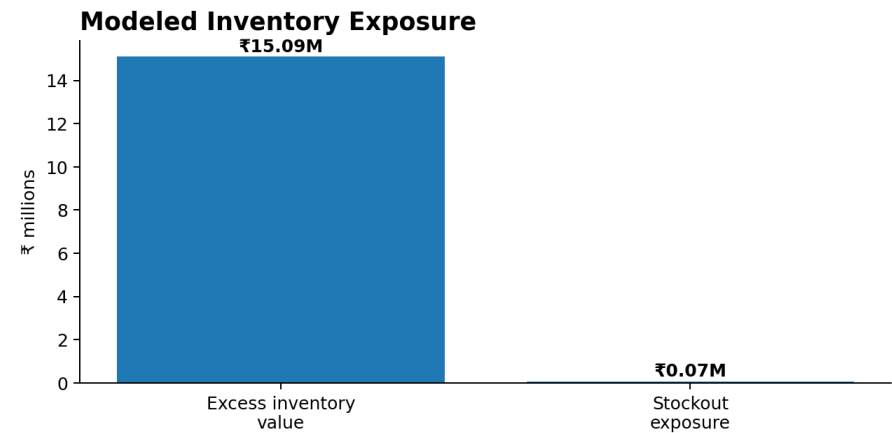
- 5 SKUs — REORDER**
Immediate replenishment candidates under the defined stockout rule.
- 21 SKUs — OVERSTOCK**
Candidates for inventory review, demand validation, or markdown planning.
- 24 SKUs — MONITOR**
No threshold-triggered action under the current rules.

Project assumption: Overstock was defined as $\text{Current Stock} > \text{Forecast} \times 4$. This threshold is a modeling/business-rule assumption and should be reviewed with the client before operational use.

03 | Business & Rupee Impact

Financial values translate the inventory flags into modeled exposure using SKU cost data.

■15.09M	■67.56K	■15.16M	4,909
Excess inventory value	Stockout exposure	Combined modeled exposure	Estimated excess units



Business interpretation

The largest modeled exposure comes from excess inventory value: **■15,093,567.07** across approximately **4,909** excess units.

Modeled stockout exposure is **■67,557.84**, based on the estimated stockout gap and SKU cost.

Use of these figures: prioritize inventory review and planning. They should not be presented as realized financial losses, guaranteed savings, or causal estimates.

04 | Actions, Workflow & Limitations

A practical handoff from analytics to inventory planning.

1. REORDER — 5 SKUs	2. OVERSTOCK — 21 SKUs
Review lead times, available stock and forecast demand; validate replenishment quantities before placing orders.	Review excess units, demand assumptions, pricing/markdown options and inventory aging before taking action.
3. MONITOR — 24 SKUs	4. GOVERNANCE
Continue monitoring forecast and stock positions; re-score as new sales and inventory snapshots arrive.	Keep raw client extracts private, document assumptions, and re-run backtests when the forecasting horizon or data changes.

Project workflow

Raw extracts → Data validation & cleaning → Analysis-ready dataset → EDA → Seasonal-naive baseline → Demand forecasting → Model evaluation → Risk scoring → Decision table → Streamlit dashboard → Rupee impact → Executive readout

Key limitations

- Forecast evaluation is based on the project's selected final evaluation window.
- Negative bias indicates systematic under-forecasting in that evaluation window.
- Risk thresholds are business-rule assumptions, including the Forecast × 4 overstock threshold.
- Financial figures are modeled inventory exposure, not realized losses or guaranteed savings.
- The Streamlit dashboard remains the appropriate interface for interactive filtering and decision exploration.

Executive takeaway: The project converts SKU-level demand forecasts into operational risk categories and rupee-denominated inventory exposure, providing a structured basis for replenishment and inventory review.